

Question ID: 603c18ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment WY is 572 , and the length of segment WX is 429 . What is the value of $\tan Z$?

Answer

- A. $\frac{3}{5}$
- B. $\frac{3}{4}$
- C. $\frac{4}{5}$
- D. $\frac{4}{3}$

Correct Answer: D

Rationale

Choice D is correct. It's given that in triangle XYZ , the measure of angle X is 90° , so triangle XYZ is a right triangle. It's also given that point W lies on segment YZ , and segment WX is perpendicular to segment YZ . It follows that triangle WYX is a right triangle. Triangle WYX and triangle XYZ are right triangles with the same interior angle measure at point Y . It follows that triangle WYX and triangle XYZ are similar, where angles W , Y , and X in triangle WYX correspond to angles X , Y , and Z , respectively, in triangle XYZ . Since corresponding angles in similar triangles are congruent, their tangents have equal value. Thus, the tangent of angle Z is equal to the tangent of angle X in triangle WYX . The opposite side of angle X in triangle WYX is segment WY , and the adjacent side is segment WX . It's given that the length of segment WY is 572 and the length of segment WX is 429 . It follows that the tangent of angle X in triangle WYX is $\frac{572}{429}$, or $\frac{4}{3}$. Therefore, the value of $\tan Z$ is $\frac{4}{3}$.

Choice A is incorrect. This is the value of $\cos Z$, not $\tan Z$.

Choice B is incorrect. This is the value of $\tan Y$, not $\tan Z$.

Choice C is incorrect. This is the value of $\sin Z$, not $\tan Z$.